NetflixDB – MySQL Practice Assignment

Database Design, Sample Data and Query Questions

Objective

Create a database for a simple video streaming platform. It should store users, movies, watch history and reviews. Then write SQL queries to answer the questions given in Section 4.

Instructions

• Use MySQL (version 8.0 or above).

• Create the database and tables exactly as shown in Section 2.

• Insert the sample data given in Section 3.

• Write one SQL query for each question in Section 4. Add a short explanation under each query.

• Submit a single .sql file containing the table creation, data insertion and all answers.

1. Database Name

NetflixDB

2. Table Structure

Table: Users

Column	Data Type	Constraint
user_id	INT	Primary Key, Auto Increment
name	VARCHAR(100)	Not Null
email	VARCHAR(150)	Unique, Not Null
registration_date	DATE	Not Null
plan	ENUM('Basic','Standard','Premium')	Default 'Basic'

Table: Movies

Column	Data Type	Constraint
movie_id	INT	Primary Key, Auto Increment
title	VARCHAR(200)	Not Null
genre	VARCHAR(100)	Not Null
release_year	YEAR	Not Null
rating	DECIMAL(3,1)	Not Null (rating out of 10)

Table: WatchHistory

Column	Data Type	Constraint
watch_id	INT	Primary Key, Auto Increment
user_id	INT	Foreign Key → Users(user_id)
movie_id	INT	Foreign Key → Movies(movie_id)
watched_date	DATE	Not Null
completion_percentage	INT	Check: between 0 and 100

Table: Reviews

Column	Data Type	Constraint
review_id	INT	Primary Key, Auto Increment
movie_id	INT	Foreign Key → Movies(movie_id)
user_id	INT	Foreign Key → Users(user_id)
review_text	TEXT	
rating	DECIMAL(2,1)	Check: between 0 and 5
review_date	DATE	Not Null

3. Sample Data

Users

user_id	name	email	registration_date	plan
1	John Doe	john.doe@example.com	2024-01-10	Premium
2	Jane Smith	jane.smith@example.com	2024-01-15	Standard
3	Alice Johnson	alice.johnson@example.com	2024-02-01	Basic
4	Bob Brown	bob.brown@example.com	2024-02-20	Premium

Movies

movie_id	title	genre	release_year	rating
1	Stranger Things	Drama	2016	8.7
2	Breaking Bad	Crime	2008	9.5
3	The Crown	History	2016	8.6
4	The Witcher	Fantasy	2019	8.2
5	Black Mirror	Sci-Fi	2011	8.8

WatchHistory

watch_id	user_id	movie_id	watched_date	completion_percentage
1	1	1	2024-02-05	100
2	2	2	2024-02-06	80
3	3	3	2024-02-10	50
4	4	4	2024-02-15	100
5	1	5	2024-02-18	90

Reviews

review_id	movie_id	user_id	review_text	rating	review_date
1	1	1	Amazing storyline and great characters!	4.5	2024-02-07
2	2	2	Intense and thrilling!	5.0	2024-02-08
3	3	3	Good show, but slow at times.	3.5	2024-02-12
4	4	4	Fantastic visuals and acting.	4.8	2024-02-16

4. Questions

Q1. List all users subscribed to the Premium plan.

Q2. Retrieve all movies in the Drama genre with a rating higher than 8.5.

Q3. Find the average rating of all movies released after 2015.

Q4. List the names of users who have watched the movie Stranger Things along with their completion percentage.

Q5. Find the name of the user(s) who rated a movie the highest among all reviews.

Q6. Calculate the number of movies watched by each user and sort by the highest count.

Q7. List all movies watched by John Doe, including their genre, rating, and his completion percentage.

Q8. Update the movie's rating for Stranger Things.

Q9. Remove all reviews for movies with a rating below 4.0.

Q10. Fetch all users who have reviewed a movie but have not watched it completely (completion percentage < 100).

Q11. List all movies watched by John Doe along with their genre and his completion percentage.

Q12. Retrieve all users who have reviewed the movie Stranger Things, including their review text and rating.

Q13. Fetch the watch history of all users, including their name, email, movie title, genre, watched date, and completion percentage.

Q14. List all movies along with the total number of reviews and average rating for each movie, including only movies with at least two reviews.

Note: Questions 8 and 9 change the data. Run the questions in order.